MATH 110 — Week 2

Notes Chapters 3–4; Homework 2

§1 Independence, span and bases

The question. For $v_1 = (1, 0)$ and $v_2 = (1, 1)$, can we write $(3, 2) = av_1 + bv_2$? The coordinate equations are $a + b = 3$, $b = 2$, so $a = 1$. We want to decide both whether such coefficients exist for every target and whether they are unique.

Definition (Notes Definitions 7–8; LADR 2.4, 2.9, 2.13, 2.15, 2.26)**.** We use the field and vector-space operations from Week 1. A linear combination of a list $v_1, \dots, v_m$ is $a_1v_1 + \cdots + a_mv_m$, with $a_j \in F$. Its span, $\operatorname{span}(v_1, \dots, v_m)$, is the set of all these combinations. The empty combination is zero; the empty list spans $\{0\}$ and is independent, since there are no nonzero coefficients in an empty list. A list is ordered and may contain repeated entries; its length is its number of entries. A list is linearly independent if $\sum_{j=1}^{m} a_jv_j = 0$ implies every $a_j = 0$; otherwise it is dependent. Here $\sum$ abbreviates the displayed sum. The list spans V if its span is V, and is a basis of V if it is independent and spans V. A space is finite-dimensional if some finite list spans it; otherwise it is infinite-dimensional.

Proposition (Notes Definition 8 and combination map; LADR 2.28 (equation 2.29))

A list is a basis exactly when every vector has a unique coefficient list. The map $\Phi : F^m \to V$, $\Phi(a_1, \dots, a_m) = \sum a_jv_j$, is injective exactly when the list is independent, and surjective exactly when it spans.

Where the proof comes from

Want. Every vector must have exactly one coefficient list.

Know. Spanning gives existence. Two representations give $\sum_j (a_j - b_j)v_j = 0$ after subtraction.

Key step. Independence forces $a_j = b_j$. Conversely, uniqueness for zero compares any zero combination with the all-zero coefficient list.

Proof. 1. The image of Φ is the span by definition, giving the surjectivity statement.
2. If the list is independent and $\Phi(a) = \Phi(b)$, then $\sum(a_j - b_j)v_j = 0$, hence $a_j = b_j$ for every j.
3. If Φ is injective and $\sum a_jv_j = 0 = \Phi(0)$, then $a = 0$. Combining the two properties proves the basis assertion.

□

Proposition (Span and single-vector summands; Notes Chapter 3, LADR 2.6)

The span is the smallest subspace containing the listed vectors. It equals $Fv_1 + \cdots + Fv_m$, where $Fv = \{av : a \in F\}$. If every $v_j \neq 0$, this sum is direct exactly when the list is independent.

Where the proof comes from

Want. The smallest subspace containing the list, and a test for when its single-vector summands form a direct sum.

Key step. Every containing subspace must contain all combinations. The sum $Fv_1 + \cdots + Fv_m$ is a subspace and consists of precisely those combinations. If $v_j \neq 0$, the equation $a_j v_j = 0$ forces $a_j = 0$, so uniqueness of summands is equivalent to independence.

Proof. 1. Fv_j is a subspace because $0 = 0v_j$ and $a(bv_j) + cv_j = (ab + c)v_j$. Its sum consists precisely of the combinations.
2. Any subspace containing all v_j contains all combinations by repeated addition and scaling, proving minimality.
3. If the list is independent, $\sum a_j v_j = 0$ forces every summand zero, so the sum is direct by Week 1 Proposition 7.
4. Conversely directness forces each $a_j v_j = 0$. If $v_j \neq 0$, Week 1 HW1 P4 gives $a_j = 0$.

□

Proposition (One or two vectors, and sublists; LADR 2.16(c,d))

A one-vector list is independent exactly when its vector is nonzero. A two-vector list is independent exactly when neither vector is a scalar multiple of the other. Every sublist of an independent list is independent.

Where the proof comes from

Want. Test a short list or a sublist for independence.

Key step. In a nontrivial relation $av + bw = 0$, a nonzero coefficient lets us solve for its vector as a multiple of the other. A relation on a sublist becomes a relation on the full list by inserting zero coefficients.

Proof. 1. For $v \neq 0$, $av = 0$ implies $a = 0$ by Week 1 HW1 P4. For $v = 0$, $1v = 0$ is a nontrivial relation, so the one-vector list is dependent.
2. If $v = cw$, the relation $v - cw = 0$ is nontrivial. Conversely any nontrivial relation $av + bw = 0$ has $a \neq 0$ or $b \neq 0$; solving for the corresponding vector expresses it as a scalar multiple of the other.
3. A relation on a sublist, extended by zero coefficients at all omitted positions, is a relation on the full list. Independence of the full list forces every coefficient to be zero, including those on the sublist.

□

Example (Notes Example 10) 1. The standard list $e_1, \ldots, e_n$ in F^n has a single 1 in each successive coordinate. The equality $(a_1, \ldots, a_n) = \sum a_j e_j$ proves spanning; its jth coordinate is a_j, so a zero combination forces each $a_j = 0$. This proves independence.
2. The sequences $(1, 1, 1, \ldots)$ and $(0, 1, 2, \ldots)$ span exactly the arithmetic sequences $(a, a + d, a + 2d, \ldots)$. For a target arithmetic sequence, its first entry is a and its second is $a + d$, so d is the second entry minus the first. In a zero combination the first entry gives $a = 0$ and the second gives $a + d = 0$, hence $d = 0$. Thus these sequences are independent as well. For example $(2, 5, 8, 11, \ldots) = 2(1, 1, \ldots) + 3(0, 1, 2, \ldots)$.
3. The empty list is a basis of $\{0\}$. Any nonempty list there contains zero and is dependent, so this is its only basis.

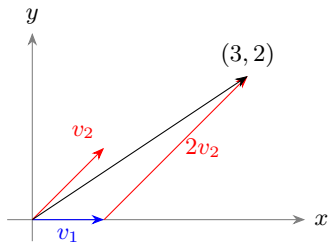
Testing a target for membership in a span: LADR 2.3 and 2.5 In $\mathbb{R}^3$, let $v = (2, 1, -3)$ and $w = (1, -2, 4)$. Solve $(r, s, t) = av + bw$:

1. The coordinate equations are $2a + b = r$, $a - 2b = s$, $-3a + 4b = t$. For $r = 17, s = -4$, the first gives $b = 17 - 2a$. Substitution in the second gives $a - 2(17 - 2a) = -4$, hence $5a = 30$ and $a = 6, b = 5$.
2. The third coordinate is then $-3 \cdot 6 + 4 \cdot 5 = 2$. Thus $(17, -4, 2) = 6v + 5w$, whereas $(17, -4, 5)$ is outside the span: its first two coordinates force $a = 6, b = 5$, which fail the third equation.

Solving only some of the equations gives candidate coefficients; substitution into every remaining equation is necessary.

Finding every dependence, not just one For $v_1 = (1,0)$, $v_2 = (1,1)$ and $v_3 = (2,1)$ in $\mathbb{R}^2$,

1. The zero combination $av_1 + bv_2 + cv_3 = 0$ gives $a + b + 2c = 0$ and $b + c = 0$. Hence $b = -c$ and $a = -c$.
2. Every relation therefore has coefficients $(-c, -c, c)$. Choosing $c = -1$ gives $v_1 + v_2 - v_3 = 0$, so the three-vector list is dependent.
3. For the first two vectors alone, set $c = 0$; the equations force $a = b = 0$. Moreover $(x, y) = (x - y)v_1 + yv_2$, so they form a basis of $\mathbb{R}^2$.



$(3, 2) = v_1 + 2v_2$. For a general target $(x, y) = av_1 + bv_2 = (a + b, b)$, the second coordinate gives $b = y$, then the first gives $a = x - y$. These coefficients exist and are unique.

Example (LADR 2.16(a), 2.18: independence and a parameter) 1. In F^4, $a(1,0,0,0) + b(0,1,0,0) + c(0,0,1,0) = (a,b,c,0)$. A zero result forces $a = b = c = 0$, so this list is independent, though it does not span the fourth coordinate.

2. Over $\mathbb{R}$ or $\mathbb{C}$, let $u = (2,3,1)$, $v = (1,-1,2)$ and $w_c = (7,3,c)$. A relation $au+bv+dw_c = 0$ gives $2a+b+7d = 0$ and $3a - b + 3d = 0$, hence $5a + 10d = 0$, $a = -2d$, $b = -3d$.
3. Its last coordinate is then $a + 2b + cd = (c - 8)d = 0$. If $c \neq 8$, this forces $d = 0$ and then $a = b = 0$. If $c = 8$, $2u + 3v - w_8 = 0$ is nontrivial. Thus dependence occurs exactly for $c = 8$.

Shortcuts

- A list containing zero or a repeated vector is dependent: write a relation with coefficient 1 on zero, or coefficients $1, -1$ on equal entries.
- To test independence, equate the combination to zero and compare coordinates; to test spanning, equate it to a general target.
- A unique solution for every target proves both properties simultaneously by the combination-map criterion.

Traps

- Nonzero vectors need not be independent: $(1,0), (2,0)$ are dependent over $\mathbb{R}$.
- The statement about direct sums requires the nonzero hypothesis: the one-term sum $F0 = \{0\}$ is direct, but the list (0) is dependent.

§2 Deleting, replacing and extending vectors

The question. Start with $(1,2), (3,6), (4,7), (5,9)$ in $\mathbb{R}^2$. Which entries can be deleted without changing the span? Here $(3,6) = 3(1,2)$ and $(5,9) = (1,2) + (4,7)$. The remaining pair is a candidate basis.

Proposition (Notes Proposition 8; LADR 2.19)
For a dependent list $v_1, \ldots, v_m$, some v_j belongs to $\operatorname{span}(v_1, \ldots, v_{j-1})$. Deleting any such v_j preserves the span.

Where the proof comes from

Want. Express one entry using earlier entries, so it can be deleted without changing the span.

Know. Dependence gives $\sum_k a_k v_k = 0$ with some $a_k \neq 0$.

Key step. Choose the last nonzero coefficient a_j. Solving for v_j then uses only earlier vectors. Substitute that expression whenever v_j occurs in a combination.

Proof. 1. Choose $\sum a_k v_k = 0$ nontrivially and let j be the largest index with $a_j \neq 0$. Then $v_j = -\sum_{k<j}(a_k/a_j)v_k$.

2. More generally, if $v_j = \sum_{k<j} c_k v_k$, an arbitrary combination becomes

$$\sum_{k=1}^{m} b_k v_k = \sum_{k<j}(b_k + b_j c_k)v_k + \sum_{k>j} b_k v_k.$$

It uses no v_j, so the original span is contained in the shortened span. This also covers $j = 1$: then $v_1 = 0$ and its term contributes nothing.

3. The shortened span is contained in the original span because its vectors already occur there; thus they are equal.

□

Proposition (Notes Proposition 9; LADR 2.30–2.31)

Every finite spanning list can be reduced to a basis by deleting vectors. Consequently every finite-dimensional space has a basis.

Where the proof comes from

Want. Make a spanning list independent while preserving its span.

Key step. Proposition 8 supplies a vector that can be deleted without changing the span. Repeat only while the list is dependent. Its finite length decreases each time, so the process ends at an independent spanning list.

Proof. 1. While the list is dependent, delete a vector supplied by Proposition 8. The span remains V.

2. At most the original number of entries can be deleted. The procedure stops with an independent spanning list, which is a basis.

□

Carrying out basis extraction: the example preceding LADR 2.30 Let $v_1 = (1, 2)$, $v_2 = (3, 6)$, $v_3 = (4, 7)$ and $v_4 = (5, 9)$ over $\mathbb{R}$.

1. Delete $v_2 = 3v_1$. Next, solving $av_1 + bv_3 = v_4$ gives $a + 4b = 5$, $2a + 7b = 9$. Subtracting twice the first equation gives $-b = -1$, so $b = 1, a = 1$. Delete $v_4 = v_1 + v_3$.
2. For the remaining pair, $av_1 + bv_3 = 0$ gives $a + 4b = 0$, $2a + 7b = 0$. Subtracting twice the first equation gives $-b = 0$, then $a = 0$; they are independent.
3. To see spanning directly, a general target (x, y) requires $a + 4b = x$, $2a + 7b = y$. Thus $b = 2x - y$, $a = 4y - 7x$, so every target has a representation. The initial list and final pair both span $\mathbb{R}^2$.

Finding the earliest removable entry: LADR 2.21 For $(1, 2, 3), (6, 5, 4), (15, 16, 17), (8, 9, 7)$ in $\mathbb{R}^3$:

1. The first vector is nonzero. The second is not its multiple: the first coordinate would give multiplier 6, but the second would then be $12 \neq 5$.
2. For the third, $a + 6b = 15$ and $2a + 5b = 16$ give $-7b = -14$, hence $b = 2, a = 3$. The last coordinate checks: $3a + 4b = 9 + 8 = 17$.
3. Thus the first removable entry is the third, because $(15, 16, 17) = 3(1, 2, 3) + 2(6, 5, 4)$. The fourth vector's value does not affect this earliest index.

Proposition (Notes Proposition 10; LADR 2.22)

If $e_1, \dots, e_m$ is independent and $f_1, \dots, f_n$ spans the same space, then $m \le n$.

Where the proof comes from

Want. Insert m prescribed independent vectors into an n-entry spanning list.

Know. The next e_k can be expressed using the current spanning list, but not using only $e_1, \dots, e_{k-1}$.

Key step. Its expression must use a remaining f_j with nonzero coefficient. Solve for that f_j and replace it by e_k; the new list still spans. After n replacements there are no f's left, so a further independent e is impossible.

Proof. 1. Suppose $e_1, \dots, e_{k-1}, f_k, \dots, f_n$ spans after relabelling unused f's. Write e_k as its combination.

2. Some coefficient of $f_k, \dots, f_n$ is nonzero; otherwise e_k lies in the span of preceding e's, contrary to independence.

3. Write $e_k = \sum_{i<k} a_i e_i + \sum_{j=k}^{n} b_j f_j$, relabelling the remaining f's so $b_k \neq 0$. Then

$$f_k = b_k^{-1} \left(e_k - \sum_{i<k} a_i e_i - \sum_{j>k} b_j f_j \right).$$

The replacement list expresses the deleted f_k and already contains all other entries, so still spans.

4. Starting with the original f list repeats this process. If $m > n$, the first n replacements make $e_1, \dots, e_n$ span, contradicting independence when e_{n+1} is expressed in that span. Thus $m \le n$.

□

Proposition (Notes Proposition 11; LADR 2.32)

Every independent list in a finite-dimensional space extends to a basis. The added vectors may be selected from any specified finite spanning list.

Where the proof comes from

Want. A basis containing the given independent $e_1, \dots, e_m$.

Know. Proposition 10 gives a spanning list with these e's first and some original f's after them.

Key step. Any dependence must use an f, since the e's are independent. The last nonzero coefficient is therefore on an f. Delete it using Proposition 8 and repeat; every e remains.

Proof. 1. The preceding construction gives a spanning list consisting first of all the e's and then some f's.

2. A dependence relation must have a nonzero coefficient on an f, since the e's alone are independent. Taking the last nonzero coefficient removes an f by Proposition 8.

3. Repeat until independent; spanning persists and no e has been removed.

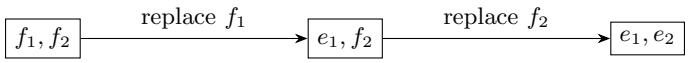
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Example (Notes Example 11)

No three vectors in F^2 are independent because $(1,0), (0,1)$ spans and Proposition 10 bounds every independent list by 2. Thus three lines in F^2 cannot form a direct sum: choosing a nonzero vector on each would give an independent three-vector list by the single-vector-summand criterion, which is impossible.

Two replacements with their equations Over $\mathbb{R}$, start from $f_1 = (1,0), f_2 = (0,1)$ and prescribe $e_1 = (1,1), e_2 = (1,-1)$.

1. $e_1 = f_1 + f_2$ gives $f_1 = e_1 - f_2$, so replace f_1 by e_1.
2. In the new list, $e_2 = e_1 - 2f_2$, hence $f_2 = (e_1 - e_2)/2$. Replace f_2 by e_2. Each old vector can still be expressed, so each displayed list spans.
3. A relation $ae_1 + be_2 = (a+b, a-b) = 0$ gives $a + b = a - b = 0$, hence $a = b = 0$. The final pair is a basis.



$f_1 = e_1 - f_2$ and $f_2 = (e_1 - e_2)/2$ express each deleted vector using the new list. Each list therefore spans the same plane.

Example (The example following LADR 2.32)

Over $\mathbb{R}$ or $\mathbb{C}$, extend $u = (2, 3, 4), v = (9, 6, 8)$ using coordinate vectors.

1. $au + bv = 0$ gives $2a + 9b = 0$, $3a + 6b = 0$. Twice the second minus three times the first gives $-15b = 0$, so $b = a = 0$.
2. The candidate $(1, 0, 0)$ is already in their span: $-\frac{2}{5}u + \frac{1}{5}v = (1, 0, 0)$. It adds nothing.
3. The candidate $(0, 1, 0)$ is outside their span: $au + bv = (0, 1, 0)$ would give $3a + 6b = 1$ and $4a + 8b = 0$, but multiplying the first by 4 and the second by 3 gives equal left sides and unequal right sides 4 and 0.
4. Thus $u, v, (0, 1, 0)$ is independent. The three coordinate equations for $(x, y, z) = au + bv + c(0, 1, 0)$ give $b = (2x - z)/10$, $a = (9z - 8x)/20$, $c = y - 3z/4$, so it spans as well.

Example (LADR 2.23–2.24: length alone can decide failure)

The standard list of length 3 spans $\mathbb{R}^3$, so Proposition 10 makes any four-vector list there dependent, including $(1, 2, 3), (4, 5, 8), (9, 6, 7), (-3, 2, 8)$. The standard list of length 4 is independent in $\mathbb{R}^4$, so no three-vector list spans it, including $(1, 2, 3, -5), (4, 5, 8, 3), (9, 6, 7, -1)$.

Shortcuts

- Extract a basis from a finite spanning list by deleting vectors expressible in preceding ones; the dependence lemma preserves the span.
- Extend an independent list by appending candidates only when they lie outside the current span; a relation involving the new vector would contradict that test.
- A longer list than a known spanning list must be dependent, without solving any coefficients.

Traps

- Removing an arbitrary vector can destroy spanning: deleting $(0, 1)$ from the standard basis of F^2 does so.
- Adding arbitrary vectors need not extend an independent list: appending $(2, 0)$ to $(1, 0)$ over $\mathbb{R}$ creates dependence.

§3 Infinite lists and polynomial spaces

The question. Can the coordinate sequences $e_1, e_2, \ldots$ produce $(1, 1, 1, \ldots)$ by a finite linear combination? Even $2e_1 - e_3 = (2, 0, -1, 0, \ldots)$ is zero past its largest selected index. We must distinguish an infinite set of available vectors from an allowed infinite sum.

Definition (Notes Definition 9; polynomial notation: LADR 2.10–2.12)**.** A set of vectors is independent if every finite subset is independent, spans V if every vector is a finite combination of members, and is a basis if both hold. No infinite summation or limiting process is part of these definitions. Write $\mathcal{P}(F)$ for formal polynomials over F, with coefficientwise addition and the usual multiplication by scalars. Its elements are finite expressions $\sum_{k=0}^{n} a_k x^k$, with equality defined by equality of coefficients. Write $\mathcal{P}_n(F)$ for those of degree at most n, including zero; degree means the largest exponent with nonzero coefficient for a nonzero polynomial. Set $\deg 0 = -\infty$, where this symbol is smaller than every integer, so zero belongs to every $\mathcal{P}_n(F)$. These are formal polynomials as in Week 1. LADR restricts F to $\mathbb{R}$ or $\mathbb{C}$ and identifies them with their functions; over a general field we use coefficients to define equality.

Example (Notes Examples 12–13) 1. $\{1, x, x^2, \ldots\}$ is a basis of $\mathcal{P}(F)$: every polynomial has finitely many coefficients, and equality of polynomials is equality of those coefficients.

2. In F^∞, let e_j have 1 in position j and 0 elsewhere. A finite relation $\sum a_j e_j = 0$ gives each $a_j = 0$ by its coordinate, so the set is independent.
3. It does not span F^∞: each finite combination has only finitely many nonzero coordinates, whereas $(1, 1, \ldots)$ does not.

Degree bounds prove infinite dimensionality: LADR 2.14 Given any finite list $p_1, \ldots, p_m$ of polynomials, choose an integer $D \geq 0$ at least as large as every degree in the list. Each polynomial has zero coefficients above degree D, so every combination $\sum a_j p_j$ also has zero coefficients above degree D. The polynomial x^{D+1} has coefficient 1 there, and is outside the span. Thus $\mathcal{P}(F)$ has no finite spanning list.

coordinate	1	2	3	4	5	$\cdots$
e_1	1	0	0	0	0	$\cdots$
e_3	0	0	1	0	0	$\cdots$
$2e_1 - e_3$	2	0	-1	0	0	$\cdots$
$(1, 1, 1, \ldots)$	1	1	1	1	1	$\cdots$

The combination $2e_1 - e_3$ is zero after index 3. Every finite combination has a largest selected index and is zero after it, so none equals the all-ones sequence.

Proposition (Notes Chapter 3, optional infinite-dimensional assertions; cited)

Assuming the axiom of choice, every independent set extends to a basis, every spanning set contains a basis, and every vector space has a basis. There is an injective map from any independent set into any spanning set. These general assertions use the axiom of choice (often through Zorn's lemma), rather than the finite deletion argument; they are cited here, not needed in subsequent proofs.

Proving a space is infinite-dimensional

1. For any proposed finite spanning list of length N in F^∞, the independent list $e_1, \ldots, e_{N+1}$ violates Proposition 10.
2. Thus no finite list spans. This argument does not assume that the e_j span the entire space.

Shortcuts

- To disprove finite dimensionality, construct independent lists of arbitrarily large finite lengths; Proposition 10 rules out every finite spanning list.
- Polynomial identities can be checked coefficient by coefficient; no evaluation at infinitely many points is required.

Traps

- An infinite independent set need not be a basis, as the coordinate sequences show.
- Over a finite field, formal polynomials must not be identified with their functions: $x^2 - x$ is nonzero as a formal polynomial over $\mathbb{F}_2$ but evaluates to zero at both field elements.

§4 Dimension and basis tests

The question. The pairs $(1, 0), (0, 1)$ and $(1, 2), (3, 5)$ both describe every vector in $\mathbb{R}^2$ uniquely. Must any basis of this space have two entries? We compare two bases by using each as an independent list and the other as a spanning list.

Proposition (Notes Proposition 12; LADR 2.34)

Any two bases of a finite-dimensional vector space have equal length.

Where the proof comes from

Want. Compare the lengths m, n of two bases.

Key step. Each basis is independent and the other spans. Proposition 10 gives $m \leq n$ and, with the roles reversed, $n \leq m$.

Proof. 1. If the lengths are m, n, independence of the first and spanning of the second give $m \leq n$.
2. Swapping the roles gives $n \leq m$, hence equality.

□

Definition (Notes Definition 10; LADR 2.35)**.** The dimension $\dim V$ of a finite-dimensional space is this common basis length. In the optional infinite-dimensional setting, basis cardinality is called dimension; equality of all infinite basis cardinalities is a cited consequence of choice, not a finite-length argument.

Example (Notes Examples 14–17; LADR 2.36) 1. $\dim F^n = n$ by the standard basis; $\dim \mathcal{P}_n(F) = n+1$: every p has a unique coefficient list $a_0, \ldots, a_n$ in $p = a_0 + a_1 x + \cdots + a_n x^n$, so the $n+1$ monomials are a basis.
2. $\dim\{0\} = 0$ by its empty basis.
3. $\mathbb{C}^2$ has dimension 2 over $\mathbb{C}$ with basis $(1,0), (0,1)$. Over $\mathbb{R}$, the list $(1,0), (i,0), (0,1), (0,i)$ is a basis, because $(a+bi, c+di) = a(1,0) + b(i,0) + c(0,1) + d(0,i)$ with unique $a, b, c, d \in \mathbb{R}$. Hence its real dimension is 4.

Proposition (Notes Proposition 13; LADR 2.38, 2.42)

In a space of dimension d, an independent list of length d is a basis, and a spanning list of length d is a basis.

Where the proof comes from

Want. Prove that a length-d list has both basis properties from just one of them.

Know. An independent list extends to a basis; a spanning list reduces to a basis. Every basis has length d.

Key step. No entry can be added or deleted: that would change the length away from d. Thus the original list is already a basis.

Proof. 1. Extend the independent list to a basis. Its length is already d, so no vector can be added.
2. Reduce the spanning list to a basis. Its length is d, so none can be deleted.

□

Proposition (Notes Proposition 14; LADR 2.25, 2.37, 2.39)

If $U \subseteq V$ is a subspace and V is finite-dimensional, then U is finite-dimensional, $\dim U \leq \dim V$, and equality holds exactly when $U = V$.

Where the proof comes from

Want. Obtain a finite basis of U before comparing its length with $\dim V$.

Know. Independent vectors in U are independent in V, so Proposition 10 bounds their number by $\dim V$.

Key step. Append a vector of U outside the current span whenever one exists. Such an addition preserves independence. The bound makes the process stop, and stopping means the list spans U.

Proof. 1. Start with the empty list in U and append vectors outside its span. Independence is preserved; Proposition 10 bounds the length by $\dim V$.
2. Thus the process terminates with an independent spanning list for U, proving it has a basis and the dimension inequality.
3. If its length equals $\dim V$, Proposition 13 makes it a basis of V, so $U = V$. The converse is immediate.

□

Proposition (LADR 2.33; subspace complements)

If V is finite-dimensional and $U \subseteq V$ is a subspace, there is a subspace W with $V = U \oplus W$.

Where the proof comes from

Want. Choose W with $V = U + W$ and $U \cap W = \{0\}$.

Key step. Extend a basis of U to a basis of V, and let W be spanned by the added vectors. Grouping a basis expansion gives the sum. A vector in the intersection would have two expansions, so uniqueness forces it to be zero.

Proof. 1. Extend a basis $u_1, \dots, u_k$ of U to $u_1, \dots, u_k, w_1, \dots, w_\ell$ of V, and put $W = \operatorname{span}(w_1, \dots, w_\ell)$.
2. The full list spans V, so $U + W = V$.
3. If $v = \sum a_j u_j = \sum b_j w_j$ belongs to the intersection, subtraction gives a relation in the full basis. Every coefficient vanishes, so $v = 0$. Thus the sum is direct.

□

Example (Notes Example 18)

For $a \in \mathbb{R}$, $1, x - a, \dots, (x - a)^n$ is a basis of $\mathcal{P}_n(\mathbb{R})$.

1. Suppose $\sum_{j=0}^{n} c_j (x - a)^j = 0$ has a nonzero coefficient, and choose the largest such index k. The term $c_k (x - a)^k$ has coefficient c_k on x^k; all terms with $j < k$ have zero coefficient on x^k. The sum therefore has nonzero x^k coefficient, contradicting equality to zero.
2. The list is independent of length $n + 1 = \dim \mathcal{P}_n(\mathbb{R})$, so Proposition 13 makes it a basis.
3. Define $p^{(k)}$ as the kth derivative, with $p^{(0)} = p$, and $k! = 1 \cdot 2 \cdots k$ with $0! = 1$. We cite the elementary-calculus rule $\frac{d}{dx}(x - a)^j = j(x - a)^{j-1}$.
4. Differentiate $p(x) = \sum c_j (x-a)^j$ exactly k times. Terms with $j < k$ become zero. The $j = k$ term becomes the constant $k! c_k$. For $j > k$, a positive power of $x - a$ remains and becomes zero at $x = a$. Thus $p^{(k)}(a) = k! c_k$, so $c_k = p^{(k)}(a)/k!$.

Shifted polynomial coordinates For $p(x) = x^2 + 2x + 3$ and $a = 1$, $p(1) = 6$, $p'(1) = 4$, $p''(1)/2 = 1$, giving $p(x) = 6 + 4(x-1) + (x-1)^2$. Direct expansion verifies all coefficients.

Example (LADR 2.27(b)–(f), 2.36, 2.40: checking bases)

All numeric examples in this box use $F = \mathbb{R}$ or $\mathbb{C}$, as in the textbook.

1. $(1,2), (3,5)$ is a basis of F^2: $a + 3b = x$, $2a + 5b = y$ give $b = 2x - y$, $a = 3y - 5x$, uniquely. Adding $(4,13)$ preserves spanning but creates the relation $19(1,2) - 5(3,5) - (4,13) = 0$, so the three-vector list is not a basis.
2. $(1,2,-4), (7,-5,6)$ is independent: a zero combination gives $a + 7b = 0$, $2a - 5b = 0$, hence $-19b = 0$ and $a = b = 0$. It does not span F^3, because its length is $2 < 3$ (Proposition 10).
3. $(1,1,0), (0,0,1)$ is a basis of $\{(x,x,y) : x, y \in F\}$: the unique coefficients for (x,x,y) are x, y. Its dimension is 2.
4. For $H = \{(x,y,z) : x + y + z = 0\}$, write $(x,y,z) = (-y)(1,-1,0) + (-z)(1,0,-1)$, since $x = -y - z$. A zero combination has second and third coordinates equal to the negatives of its two coefficients; they must vanish. This is a basis of H, so $\dim H = 2$.
5. The pair $(5,7), (4,3)$ of LADR 2.40 is independent: $5a + 4b = 0$, $7a + 3b = 0$ imply $13b = 0$ by taking seven times the first minus five times the second. Then $b = a = 0$. Its length equals $\dim F^2$, so Proposition 13 proves it is a basis without another spanning calculation.
6. Another pair mentioned after LADR 2.27 is $(7,5), (-4,9)$. From $7a - 4b = 0$ and $5a + 9b = 0$, seven times the second minus five times the first gives $83b = 0$, so $a = b = 0$. Again its length makes it a basis.

Example (LADR 2.41: determining a dimension before finding all coefficients)

Let $U = \{p \in \mathcal{P}_3(\mathbb{R}) : p'(5) = 0\}$. The zero polynomial belongs, and $(ap+q)'(5) = ap'(5) + q'(5) = 0$ for $p, q \in U$, so this is a subspace.

1. The list $1, (x-5)^2, (x-5)^3$ lies in U, since the respective derivatives at 5 are $0, 2(5-5), 3(5-5)^2$. It is independent: the coefficient of x^3 in a relation is the third coefficient; after that is zero the coefficient of x^2 is the second; the constant coefficient then gives the first.
2. Thus $\dim U \geq 3$. But $x \notin U$, because its derivative at 5 is 1. Hence U is a proper subspace of the four-dimensional space $\mathcal{P}_3(\mathbb{R})$; Proposition 14 gives $\dim U < 4$.
3. Therefore $\dim U = 3$, and the independent three-member list is a basis by Proposition 13. Directly, the shifted expansion $p = c_0 + c_1(x-5) + c_2(x-5)^2 + c_3(x-5)^3$ has $p'(5) = c_1$, so the condition removes exactly the $(x-5)$ term.

Shortcuts

- Once the dimension is known, check only independence or spanning for a list of the correct length.
- Nonzero polynomials with distinct degrees are independent, because the coefficient at the highest degree with a nonzero coefficient cannot cancel.

Traps

- Always specify the field: $\mathbb{C}^2$ has two different dimensions above.
- Equal dimension forces equal nested subspaces only in finite dimension: the proper subspace of polynomials with zero constant coefficient and $\mathcal{P}(\mathbb{R})$ both have countably infinite monomial bases.

§5 Dimensions of sums and intersections

The question. For $U = \{(x,y,0)\}$ and $W = \{(x,0,z)\}$ in $\mathbb{R}^3$, each has a two-vector basis, yet $U + W = \mathbb{R}^3$ has a three-vector basis. The shared vector $(1,0,0)$ belongs to both plane bases. How do we choose and count basis vectors without duplicating the intersection?

Proposition (Notes Proposition 15)

Let U, W be finite-dimensional subspaces of a common space. A basis $v_1, \dots, v_r$ of $U \cap W$ can be extended to bases $(v_1, \dots, v_r, u_1, \dots, u_k)$ of U and $(v_1, \dots, v_r, w_1, \dots, w_\ell)$ of W. Their joined list with each v_j only once is a basis of $U + W$.

Where the proof comes from

Want. A basis of $U + W$ with the common intersection counted once.

First try. Joining arbitrary bases can repeat vectors from the intersection.

Key step. Start with a basis $v_1, \dots, v_r$ of $U \cap W$ and extend it inside each subspace. In a relation on the joined list, the U terms equal a vector of W, hence belong to the intersection. Express it with the v's and use independence in the U basis to remove the added u terms. The remaining relation is in the W basis.

Proof. 1. Extend the intersection basis in U and W by Proposition 11. The joined list spans their sum by expanding both summands.

2. In a relation $\sum a_j v_j + \sum b_j u_j + \sum c_j w_j = 0$, the first two sums equal $-\sum c_j w_j$ and belong to both U and W.
3. Write this common vector as $\sum d_j v_j$. Then $\sum (a_j - d_j) v_j + \sum b_j u_j = 0$ is a relation in the chosen U basis. Independence of that basis gives every $b_j = 0$ (and $a_j = d_j$).
4. The original relation is now in the W basis, so all remaining coefficients vanish.

□

Proposition (Notes Proposition 16; LADR 2.43)

For finite-dimensional subspaces U, W of a common space, $\dim(U + W) = \dim U + \dim W - \dim(U \cap W)$.

Where the proof comes from

Want. Count the basis from Proposition 15.

Know. The two subspace bases have lengths $r + k$ and $r + \ell$; the joined basis has length $r + k + \ell$.

Key step. Subtract the r common entries counted twice.

Proof. 1. Proposition 15 gives $\dim(U \cap W) = r$, $\dim U = r + k$, $\dim W = r + \ell$, $\dim(U + W) = r + k + \ell$.

2. Substitute to get $(r + k) + (r + \ell) - r = r + k + \ell$.

□

Example (Notes Example 19: computing the sum and intersection)

Let $u_1 = (1, 1, 0)$, $u_2 = (0, 1, 1)$, $w_1 = (1, 2, 2)$, $w_2 = (2, 2, 1)$, and $U = \operatorname{span}(u_1, u_2)$, $W = \operatorname{span}(w_1, w_2)$ in F^3. The calculations below work over any field.

1. $au_1 + bu_2 = (a, a + b, b)$ is zero only if $a = b = 0$. For $cw_1 + dw_2 = 0$, the equations are $c + 2d = 0$, $2c + 2d = 0$, $2c + d = 0$. The second minus the first gives $c = 0$; the third then gives $d = 0$. Thus both subspaces have dimension 2.
2. In a relation $au_1 + bu_2 + cw_1 = 0$, the first and third coordinates give $a = -c, b = -2c$. The middle becomes $a + b + 2c = -c = 0$, so all coefficients vanish. These three vectors form a basis of F^3 by Proposition 13; hence $U + W = F^3$.
3. The dimension formula now gives $\dim(U \cap W) = 2 + 2 - 3 = 1$. To find an actual generator, equate the two generic combinations:
$$(a, a + b, b) = (c + 2d, 2c + 2d, 2c + d).$$
The first and third give $a = c + 2d$, $b = 2c + d$. Substituting in the middle gives $3c + 3d = 2c + 2d$, so $c + d = 0$.
4. Set $c = t, d = -t$. Then $a = -t, b = t$, and the common vector is $t(-1, 0, 1)$. Conversely $(-1, 0, 1) = -u_1 + u_2 = w_1 - w_2$, so every such multiple is in both subspaces. Therefore $U \cap W = \operatorname{span}((-1, 0, 1))$.

basis of U	v	u	
basis of W	v		w
basis of $U+W$	v	u	w

In Example 19, $v=(-1,0,1)$ spans the intersection, $u=(1,1,0)$ and $w=(1,2,2)$. Since $u_2=v+u$ and $w_2=w-v$, these are bases of the two planes. The common vector v is counted once: $\dim(U+W)=2+2-1=3$.

Example (Notes Example 20)

For finite-dimensional subspaces U_1, U_2, U_3 of a common space,

1. applying Proposition 16 twice gives

$$\begin{aligned}\dim(U_1+U_2+U_3) &= \dim U_1+\dim U_2+\dim U_3\\ &\quad -\dim(U_1\cap U_2)-\dim((U_1+U_2)\cap U_3).\end{aligned}$$

2. The three lines $F(1,0)$, $F(0,1)$, $F(1,1)$ have zero pairwise intersections but their sum has dimension 2, not 3. Hence replacing the last intersection by pairwise intersections is invalid.

Proposition (Notes Proposition 17; LADR 3.94 and the comparison in 2C)

For a finite family of finite-dimensional subspaces,

$$\dim(U_1+\cdots+U_m)\le\sum_{j=1}^{m}\dim U_j,$$

with equality if and only if their sum is direct.

Where the proof comes from

Want. Compare the dimension of the sum with the total number of basis vectors from its summands.

Know. Joining bases of the U_j gives a spanning list for the sum. Its length is $N=\sum_j \dim U_j$.

Key step. Equality of dimension with this length means the joined list is a basis. A relation in it, grouped by subspace, is a zero representation $u_1+\cdots+u_m=0$. Thus independence of the joined list is precisely the direct-sum condition.

Proof. 1. Join bases of the U_j. The resulting length-N list spans the sum, so basis reduction gives $\dim(U_1+\cdots+U_m)\le N$.

2. If the sum is direct, group a relation in the joined list as $u_1+\cdots+u_m=0$, with $u_j\in U_j$. Week 1 Proposition 7 gives every $u_j=0$. Independence of each chosen basis then makes all coefficients zero. The joined list is a basis, so equality holds.

3. Conversely, equality makes the length-N spanning list a basis by Proposition 13. Expand any zero sum $u_1+\cdots+u_m=0$ in the individual bases. Independence of the joined list forces every coefficient zero, hence every $u_j=0$. Week 1 Proposition 7 gives directness.

□

The notes prove the direct case by induction instead: each U_k meets the preceding sum only at zero, so Proposition 16 gives $\dim(U_1+\cdots+U_k)=\dim(U_1+\cdots+U_{k-1})+\dim U_k$. Starting at $k=1$ gives the same sum of dimensions.

Shortcuts

- Find a sum basis by joining generators and deleting redundancy. To find an intersection, equate the two generic combinations and solve.
- If $U+W=V$ and $\dim U+\dim W=\dim V$, the formula forces a zero intersection, hence a direct sum.
- A single nonzero intersection vector gives the entire intersection when its dimension is already known to be one.

Traps

- Pairwise zero intersections do not make three summands direct: the three lines above give a counterexample.
- Do not subtract the dimension of $U \cup W$: the union need not even be a vector space (Week 1's coordinate axes).

§6 Problem set

HW2 P1 (LADR 2A.11). Prove or give a counterexample: if $v_1, \dots, v_n$ and $w_1, \dots, w_n$ are independent lists in V, then $v_1 + w_1, \dots, v_n + w_n$ is independent.
Hint: Try opposite nonzero vectors.

HW2 P2. Let F^∞ be all sequences with coordinatewise operations and e_j the jth coordinate sequence. (a) Verify the vector-space axioms. (b) Prove the space is infinite-dimensional. (c) Prove the e_j do not span it.
Hint: Use coordinatewise field laws, Proposition 10, and the finite-combination condition.

HW2 P3 (LADR 2B.5). If $V = U + W$ is finite-dimensional, prove that V has a basis whose vectors all lie in $U \cup W$.
Hint: Join bases of U, W and apply Proposition 9.

HW2 P4. (a) Every nonzero proper subspace of F^2 has form $\operatorname{span}(v)$ for some nonzero v. (b) Prove that $v, w \in F^2$ form a basis iff neither is a scalar multiple of the other.
Hint: Apply the dimension bound; in a nontrivial dependence solve for a vector with nonzero coefficient.

HW2 P5 (LADR 2C.12). Suppose $U, W \subseteq \mathbb{R}^8$ are subspaces, $\dim U = 3$, $\dim W = 5$, $U + W = \mathbb{R}^8$. Prove $\mathbb{R}^8 = U \oplus W$.
Hint: Use Proposition 16 and dimension zero.

HW2 P6. Let $U = \{p \in \mathcal{P}_4(\mathbb{R}) : p(1) = p'(1) = 0\}$. (a) Prove it is a subspace. (b) Find a basis and its dimension directly, without rank–nullity.
Hint: Expand in $1, x-1, \dots, (x-1)^4$ and evaluate the first two coefficients.

Exam practice (Practice Midterm 1, Question 1). Let $u_1 = (1,1,0,0)$, $u_2 = (0,1,1,0)$, $w_1 = (0,0,1,1)$, $w_2 = (1,0,0,-1)$, $U = \operatorname{span}(u_1, u_2)$ and $W = \operatorname{span}(w_1, w_2)$. (a) Prove $\mathbb{R}^4 = U \oplus W$. (b) Write $(2,1,3,0) = u + w$ with $u \in U, w \in W$.
Hint: Compare all four coordinates in $au_1 + bu_2 + cw_1 + dw_2$.